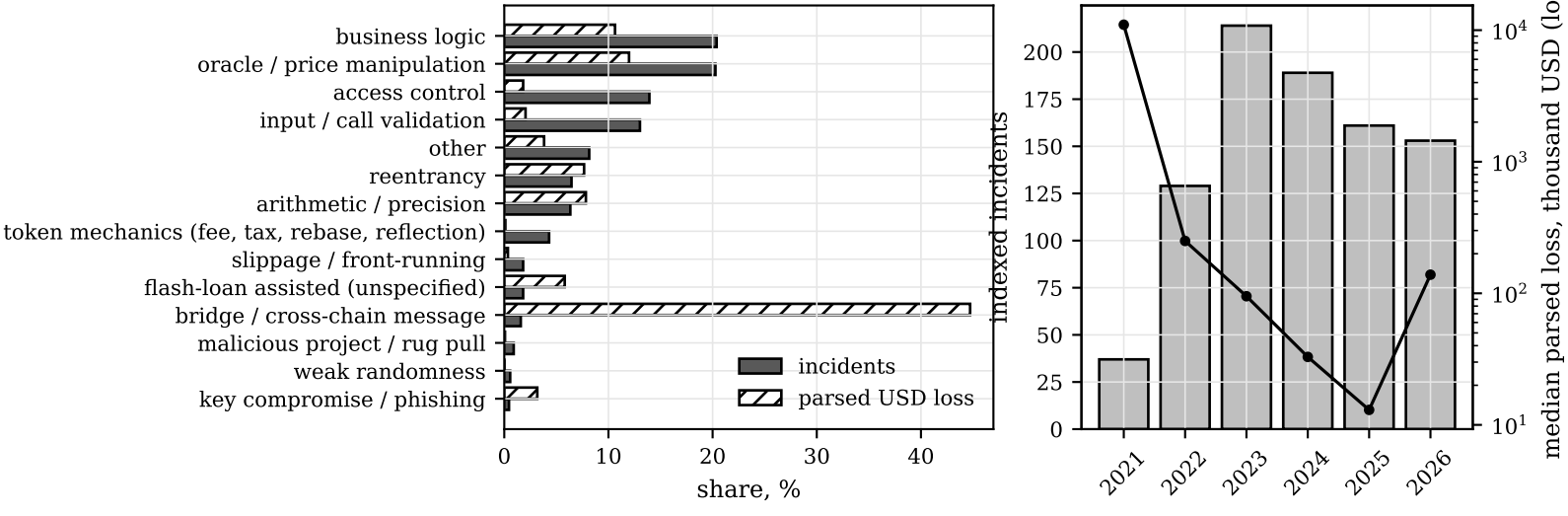
(a) root-cause classes, 2021-2026
business logic
oracle / price manipulation
access control
input / call validation
other
reentrancy
arithmetic / precision
token mechanics (fee, tax, rebase, reflection)
slippage / front-running
flash-loan assisted (unspecified)
bridge / cross-chain message
malicious project / rug pull
weak randomness
key compromise / phishing
incidents
parsed USD loss
0
10
20
30
40
share, %
(b) incidents and median loss per year
indexed incidents
0
25
50
75
100
125
150
175
200
median parsed loss, thousand USD (log)
10^1
10^2
10^3
10^4
2021
2022
2023
2024
2025
2026